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(54) **TIBIAL PLATEAU LEVELING OSTEOTOMY
PLATE**

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Publication Classification

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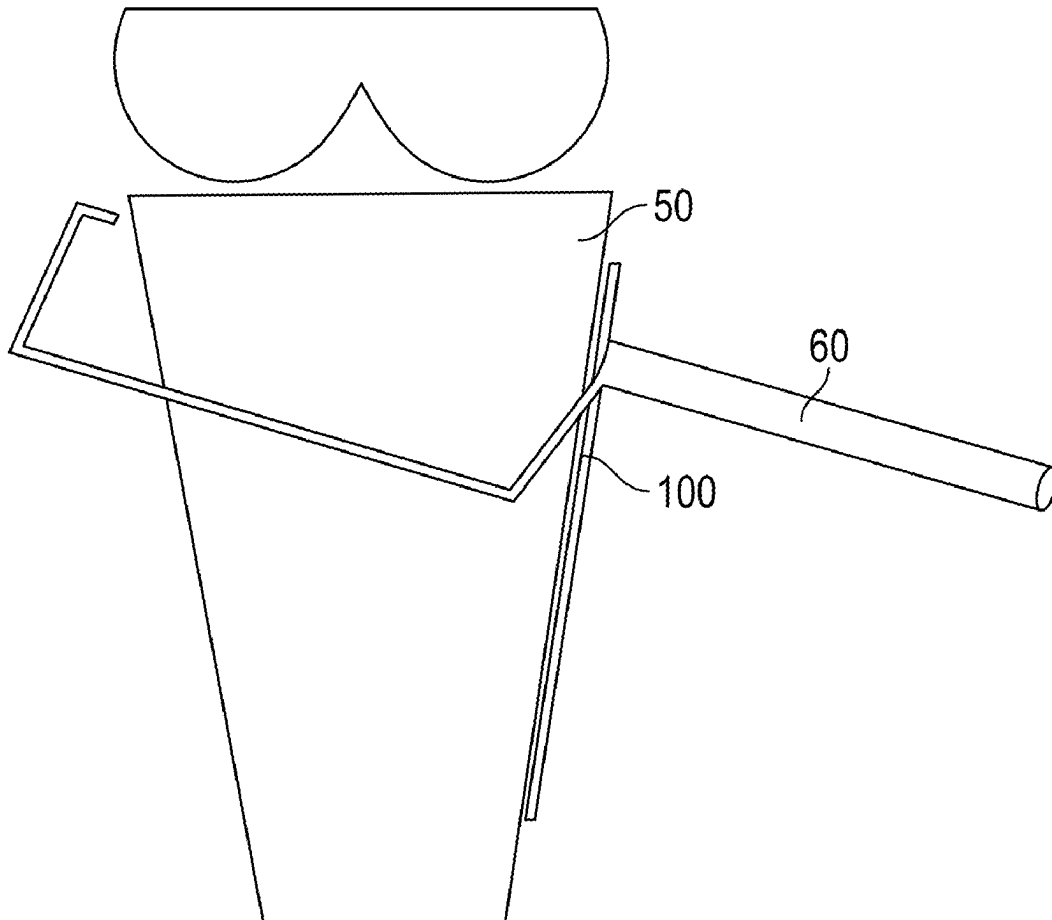
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(63) Continuation of application No. 17/858,123, filed on Jul. 6, 2022, now Pat. No. 12,329,427, which is a continuation of application No. 17/672,873, filed on Feb. 16, 2022, now Pat. No. 11,419,648, which is a continuation of application No. 17/348,888, filed on Jun. 16, 2021, now Pat. No. 11,298,167, which is a continuation of application No. 16/420,602, filed on May 23, 2019, now Pat. No. 11,096,729, which is a continuation of application No. 14/533,768, filed on Nov. 5, 2014, now Pat. No. 10,299,841.

(57) **ABSTRACT**

A TPLO plate configured to accommodate flexible strands (flexible loops) attached to tissue to be attached to bone. The TPLO plate has one or more suture eyelets, to allow the user (surgeon) to pass a flexible strand through the eyelets and attach/reattach the tissue to the plate at the anatomical location, and to improve the rotational stability of the joint or bone. The eyelets may have various shapes, forms and configurations and may be provided on or within a surface of the TPLO bone plate in any number, depending on the characteristics of the fractured bone or bone segments, or of the plate design. The eyelets preferably receive one or more flexible strands.



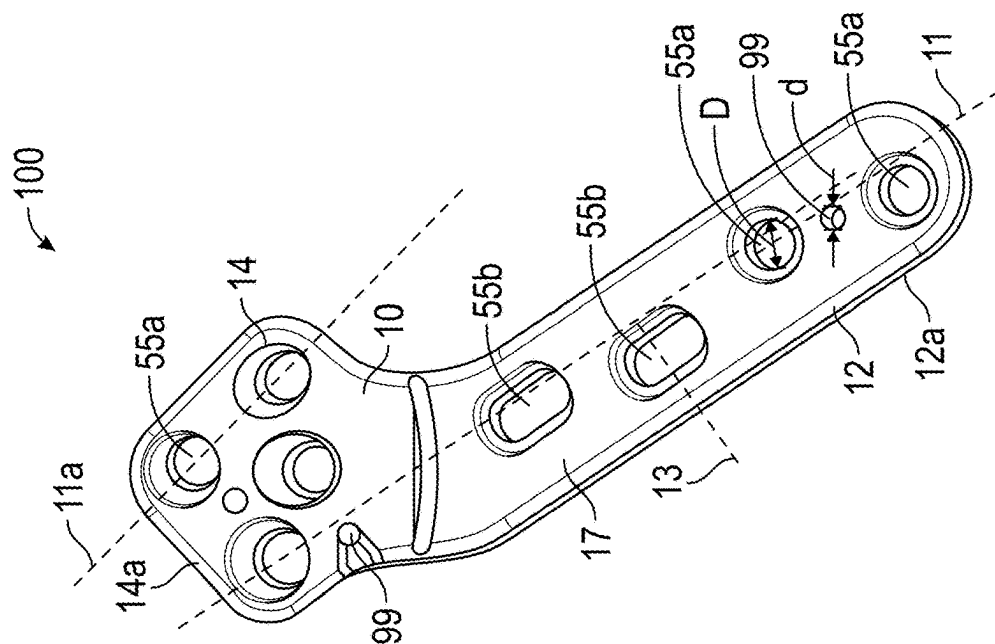


FIG. 2

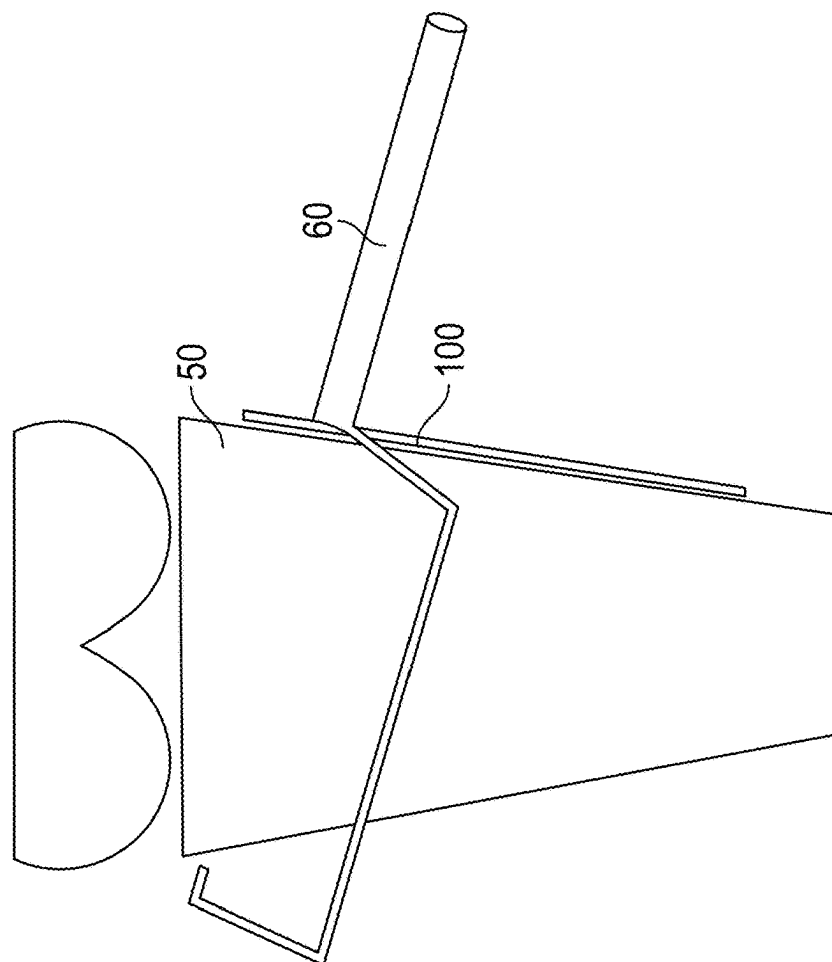
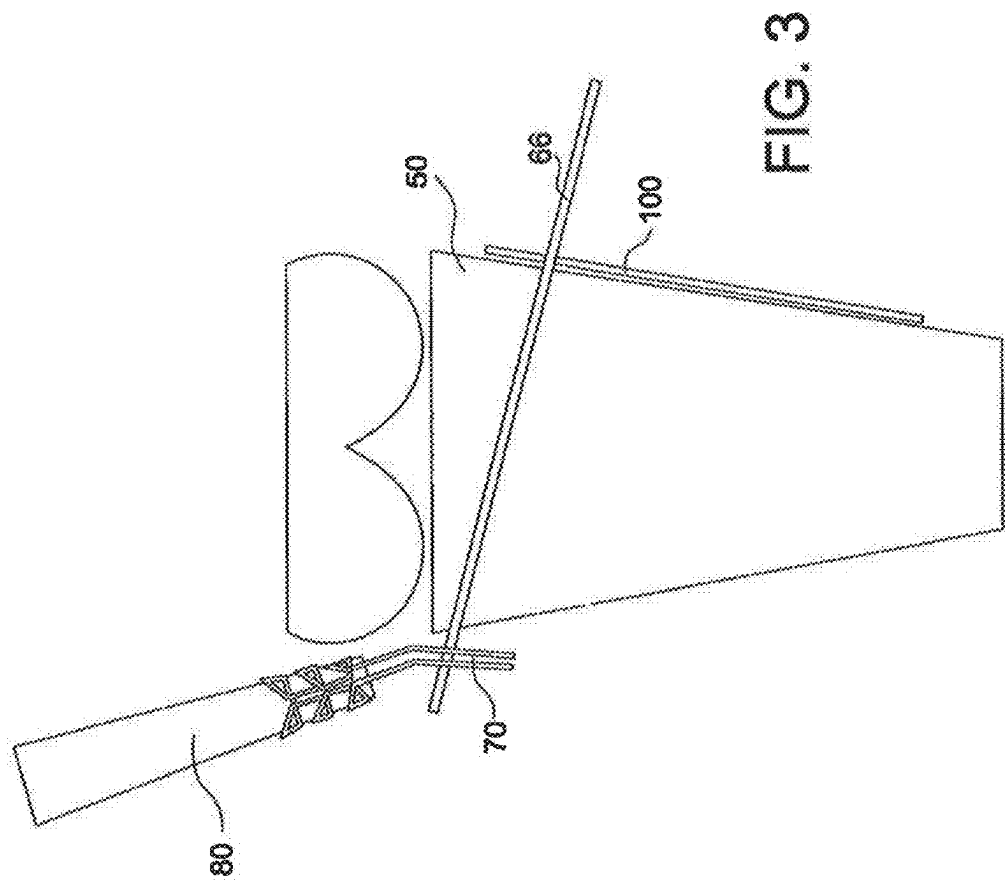


FIG.



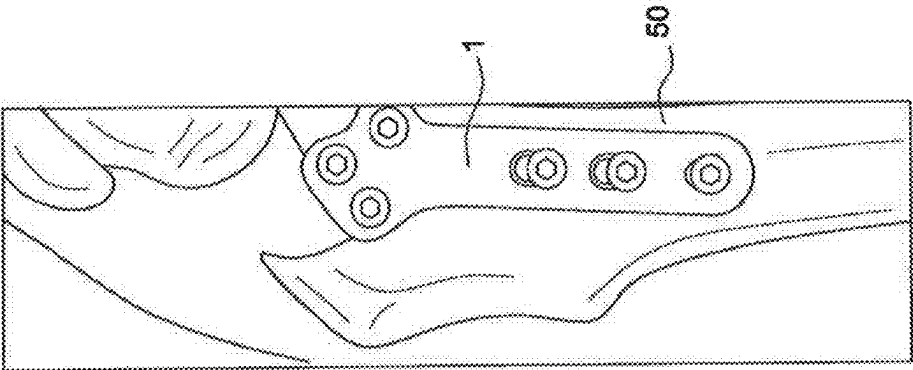


FIG. 6

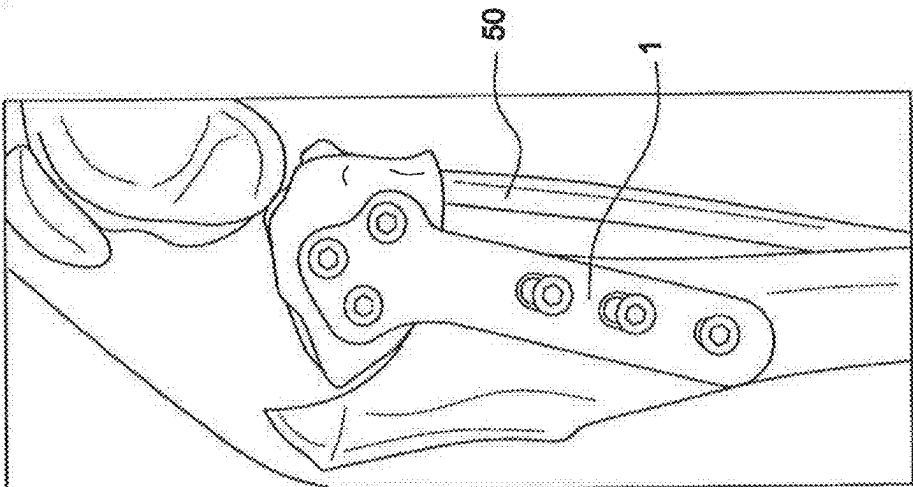


FIG. 5

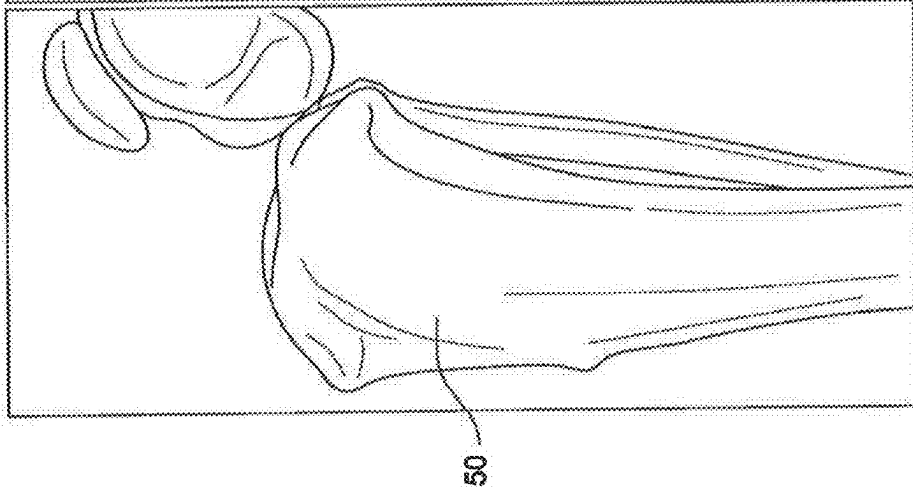
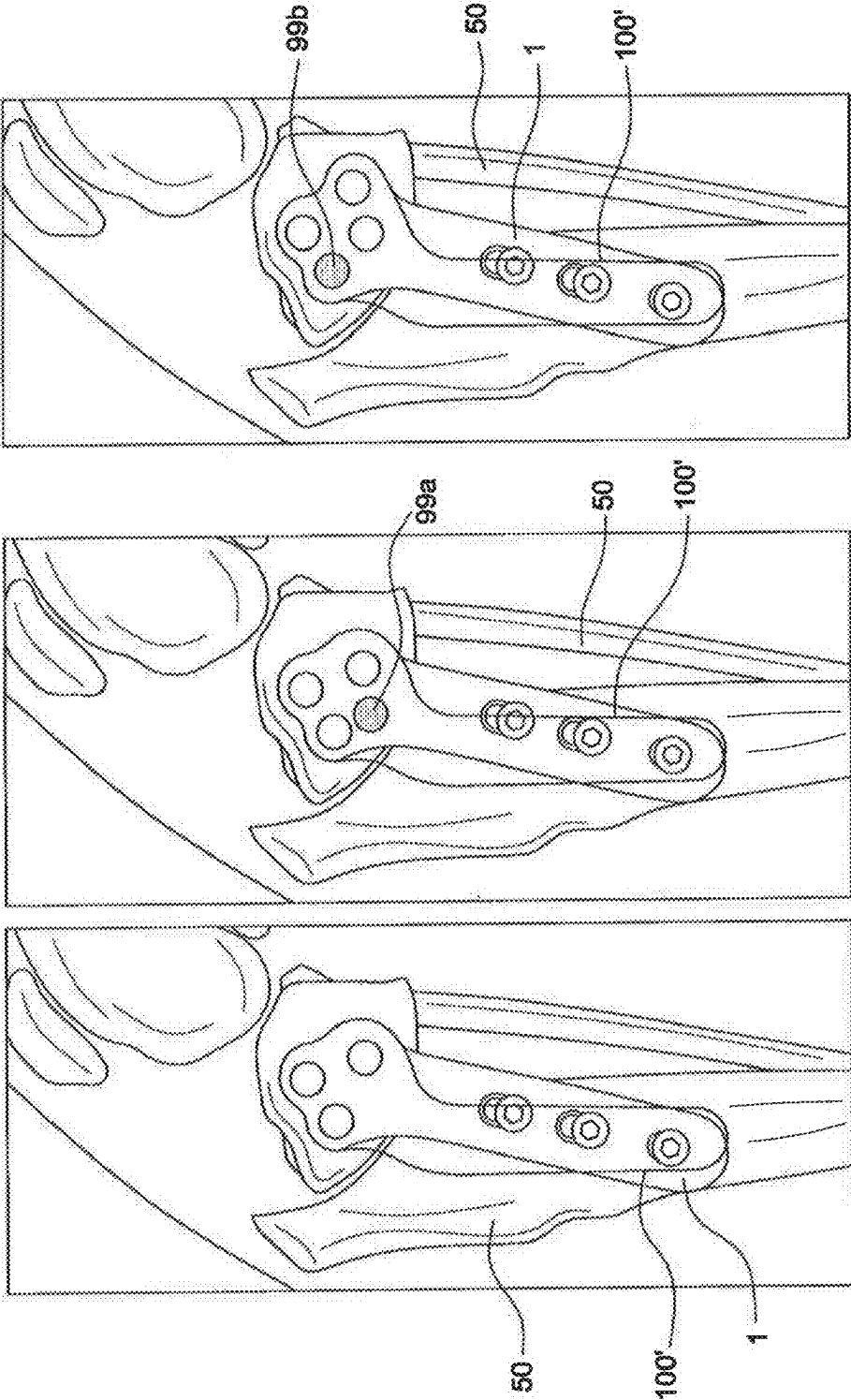


FIG. 4



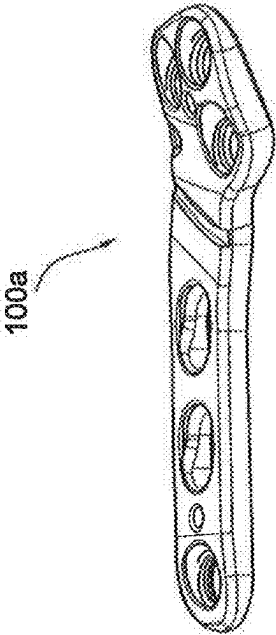


FIG. 10(a)

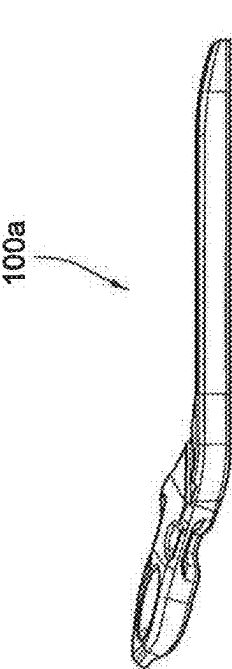


FIG. 10(b)

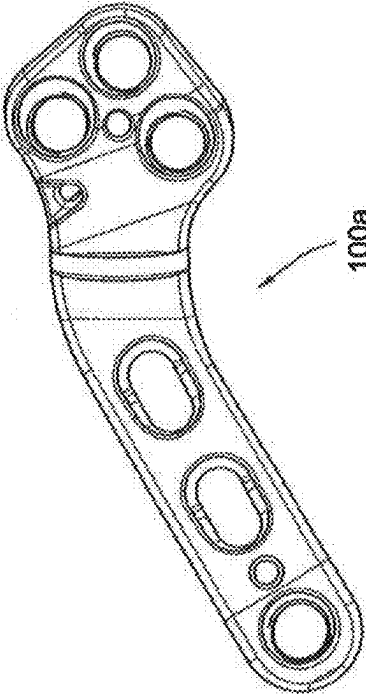


FIG. 10(c)

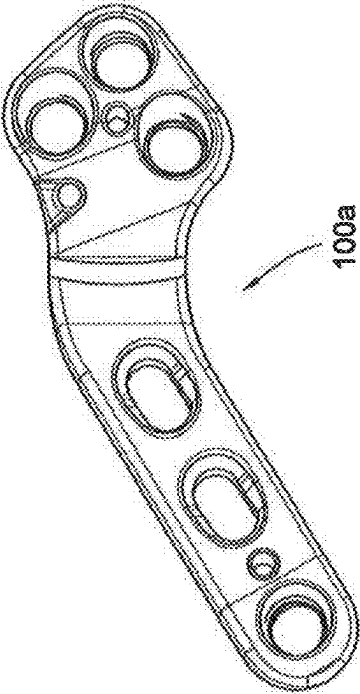


FIG. 10(d)

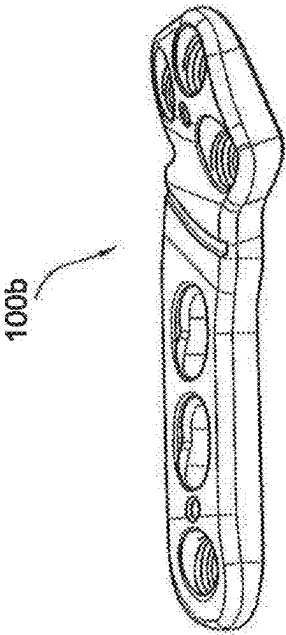


FIG. 11(a)

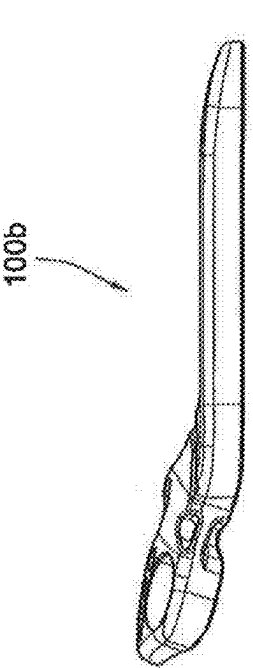


FIG. 11(b)

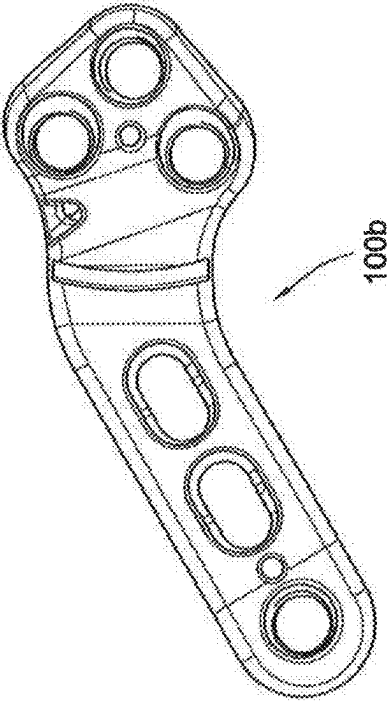


FIG. 11(c)

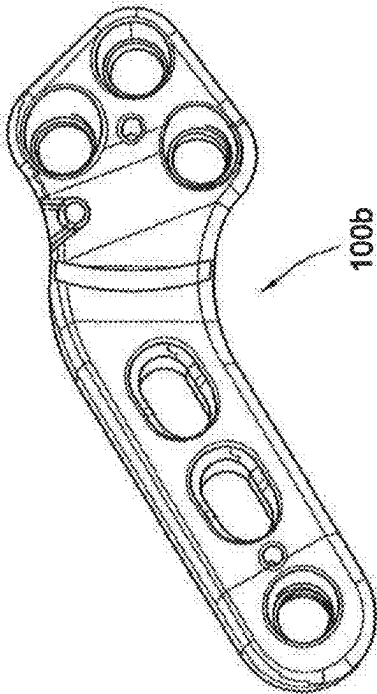


FIG. 11(d)

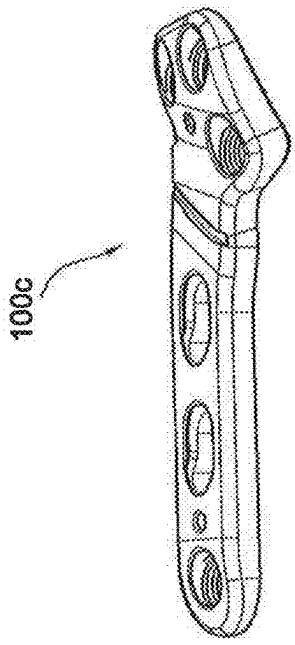


FIG. 12(a)

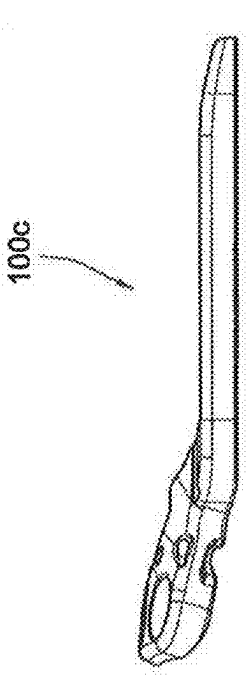


FIG. 12(b)

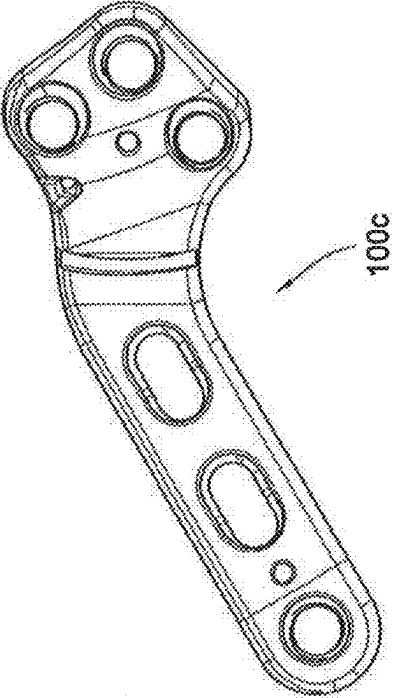


FIG. 12(c)

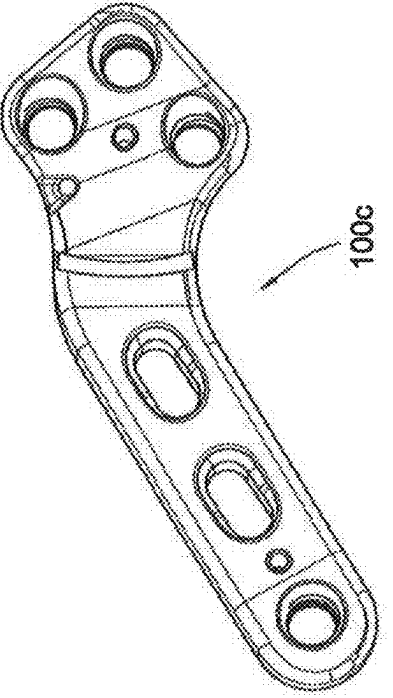


FIG. 12(d)

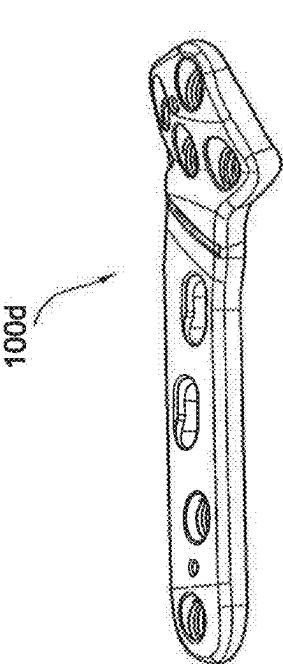


FIG. 13(a)

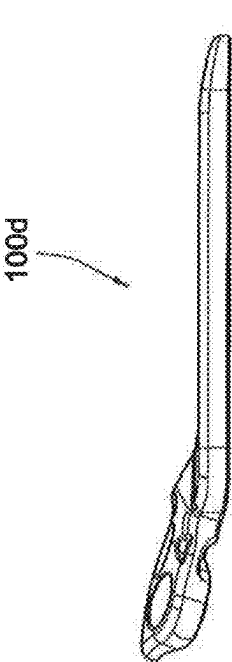


FIG. 13(b)

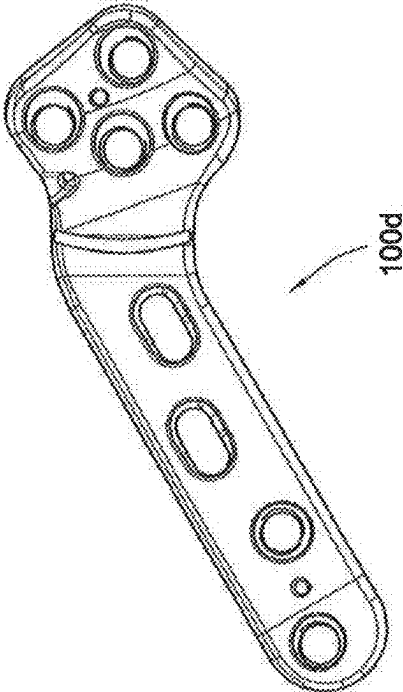


FIG. 13(c)

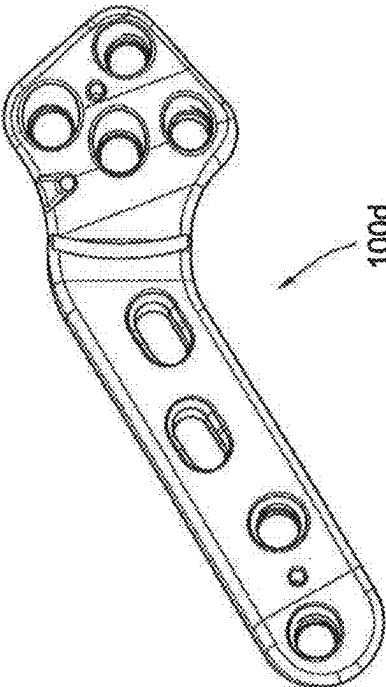


FIG. 13(d)

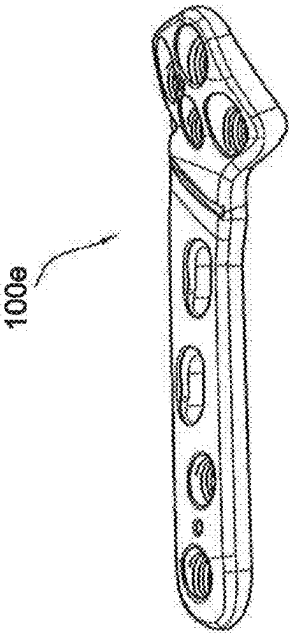


FIG. 14(a)

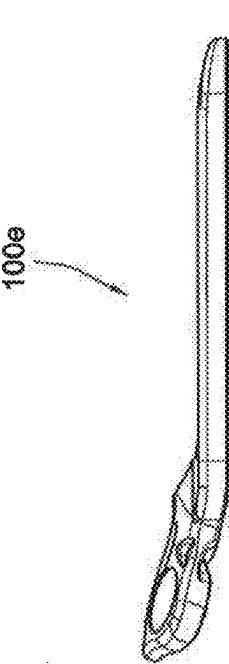


FIG. 14(b)

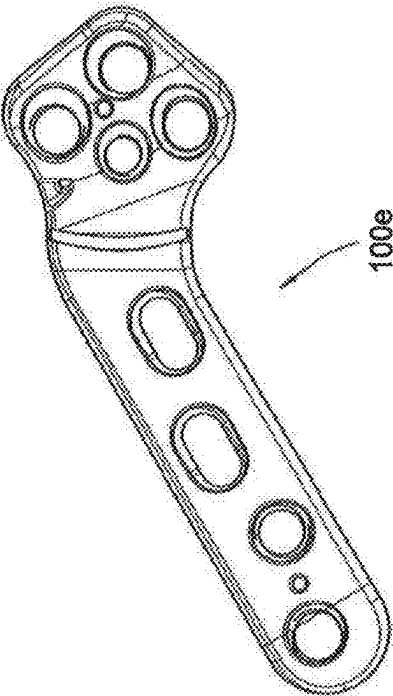


FIG. 14(c)

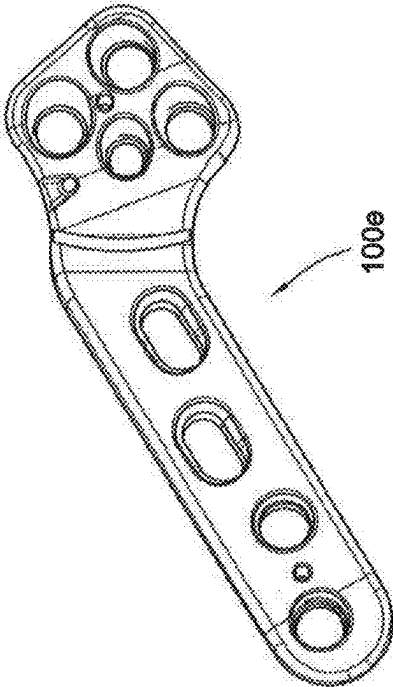


FIG. 14(d)

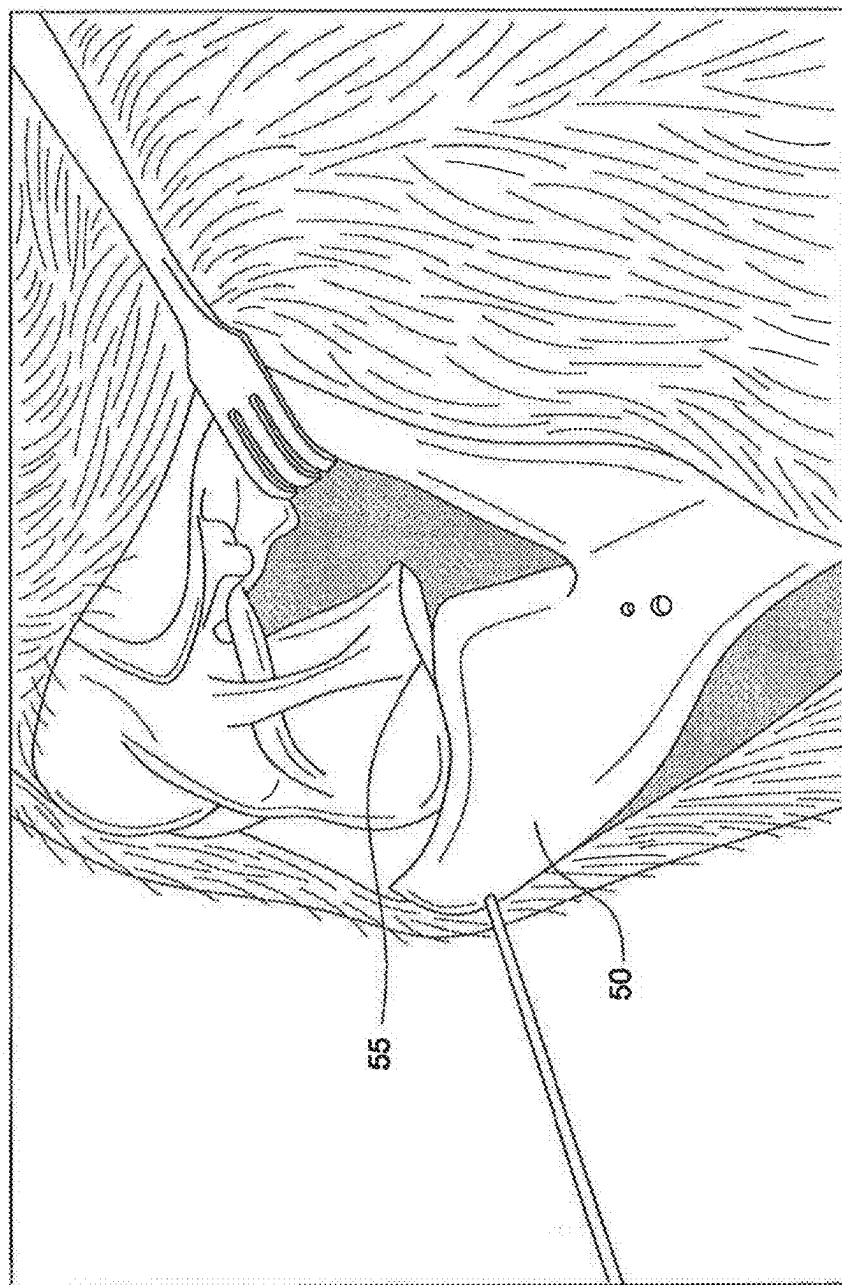


FIG. 15

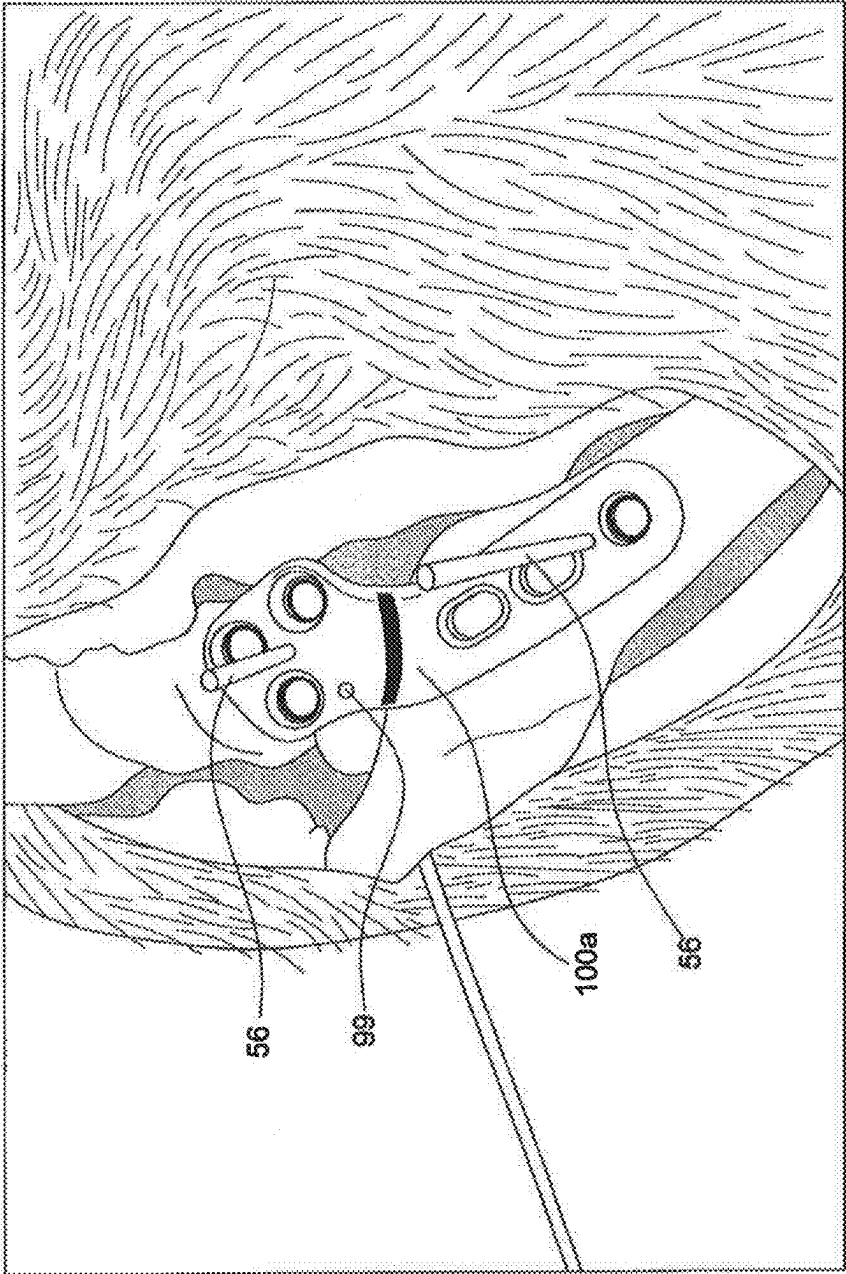


FIG. 16

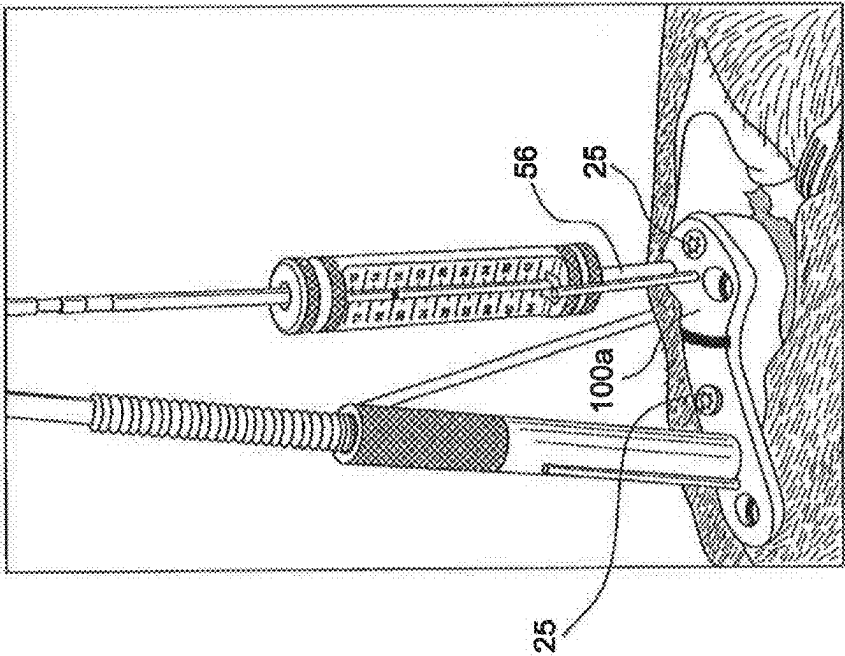


FIG. 17

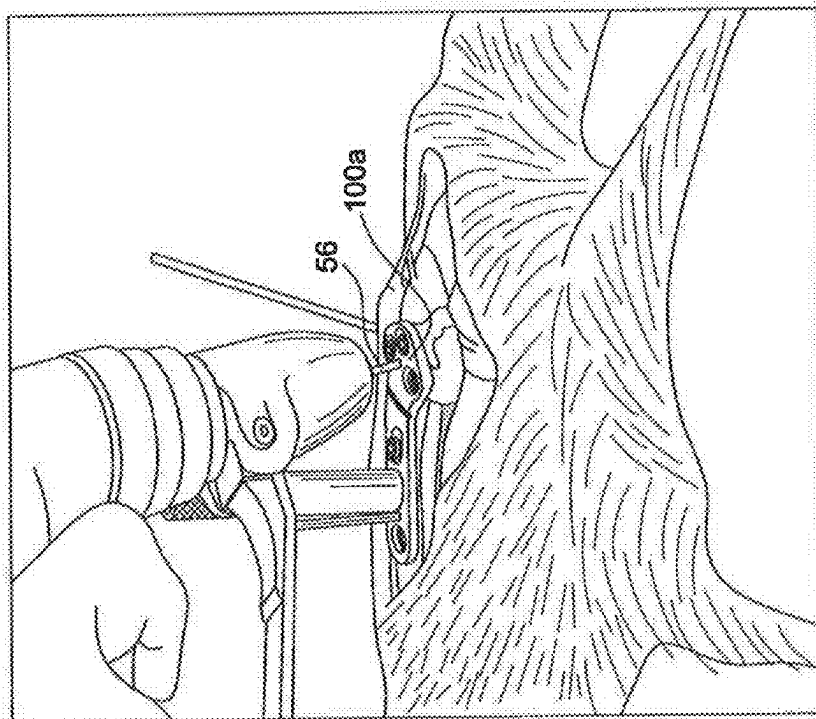


FIG. 18

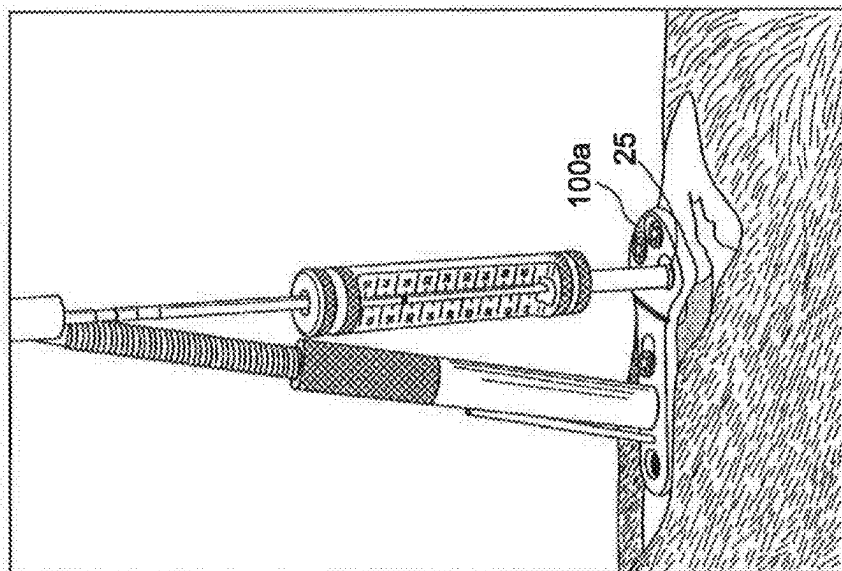


FIG. 19

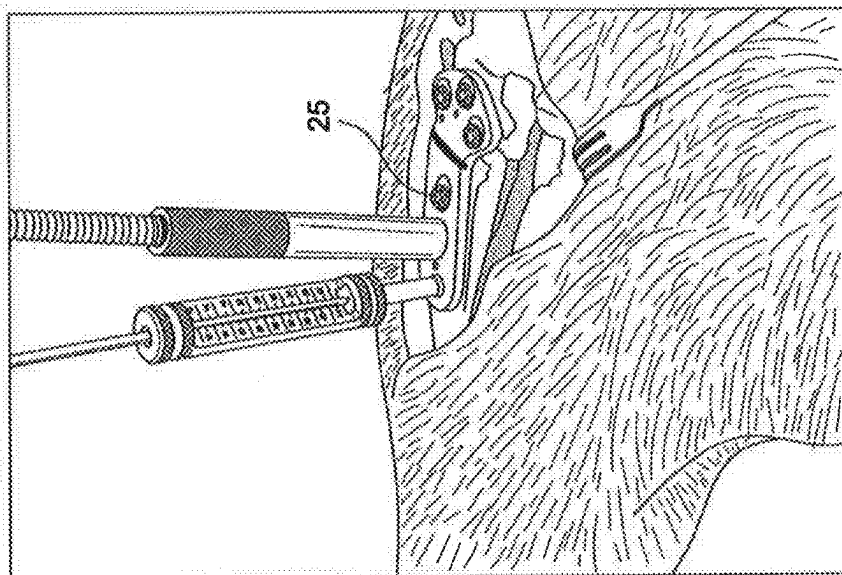


FIG. 20

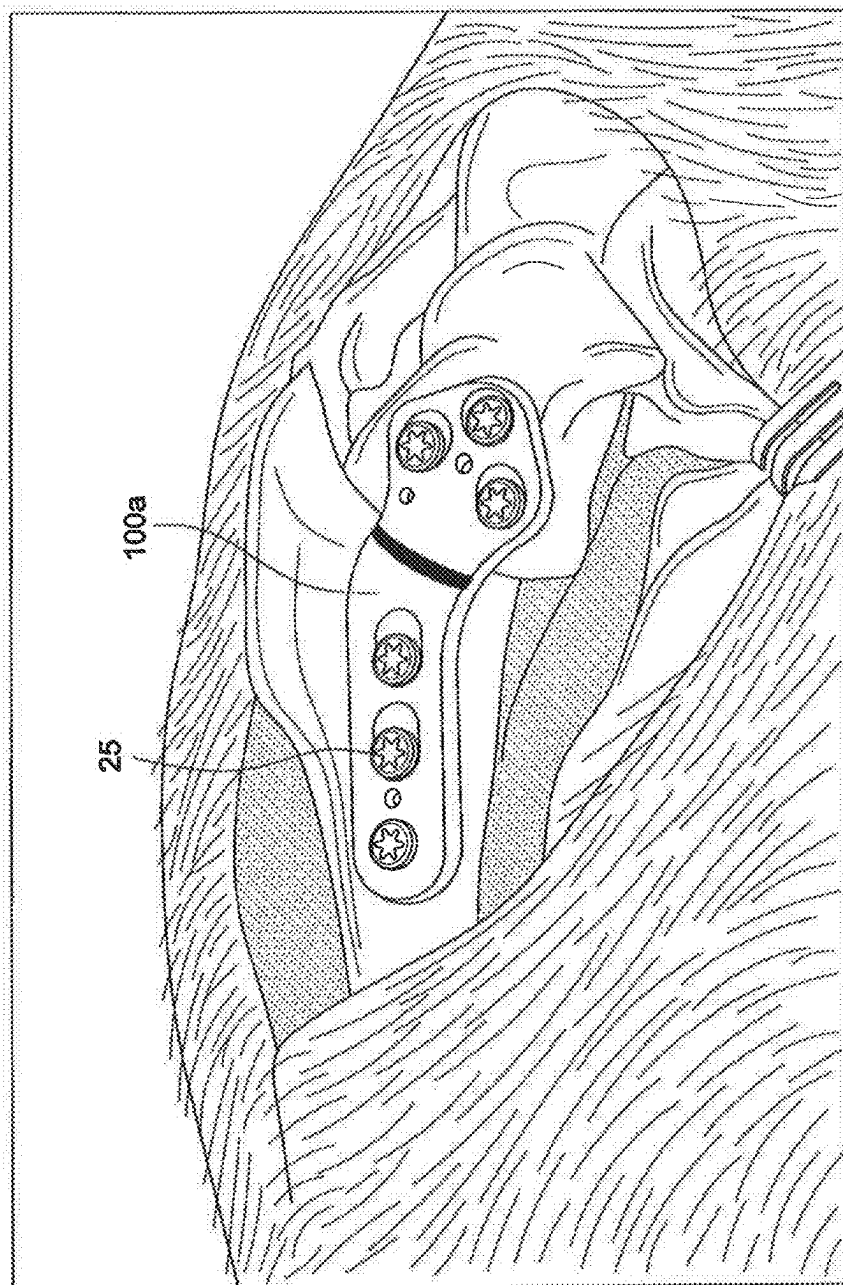


FIG. 21

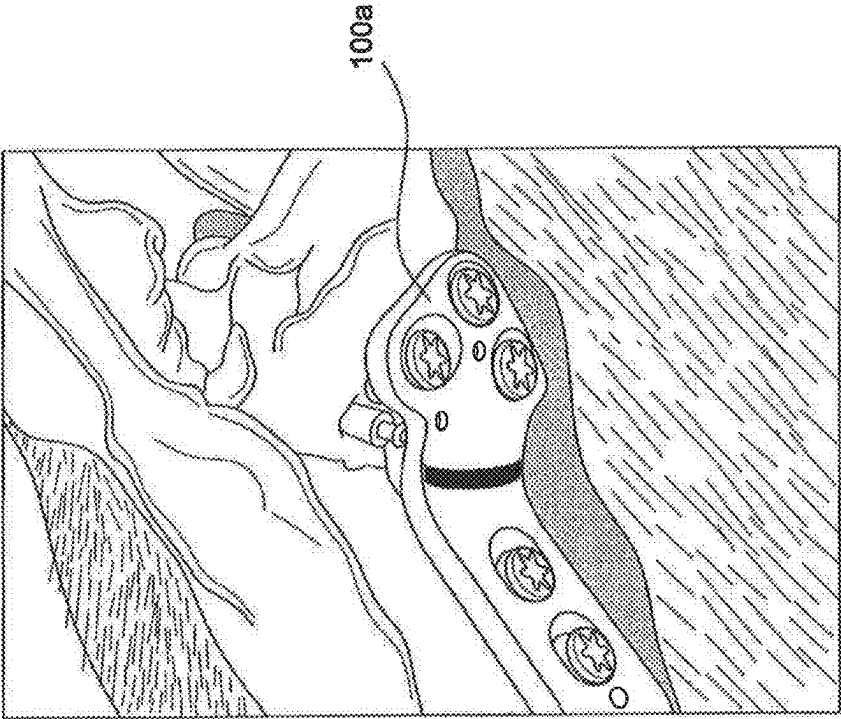


FIG. 23(b)

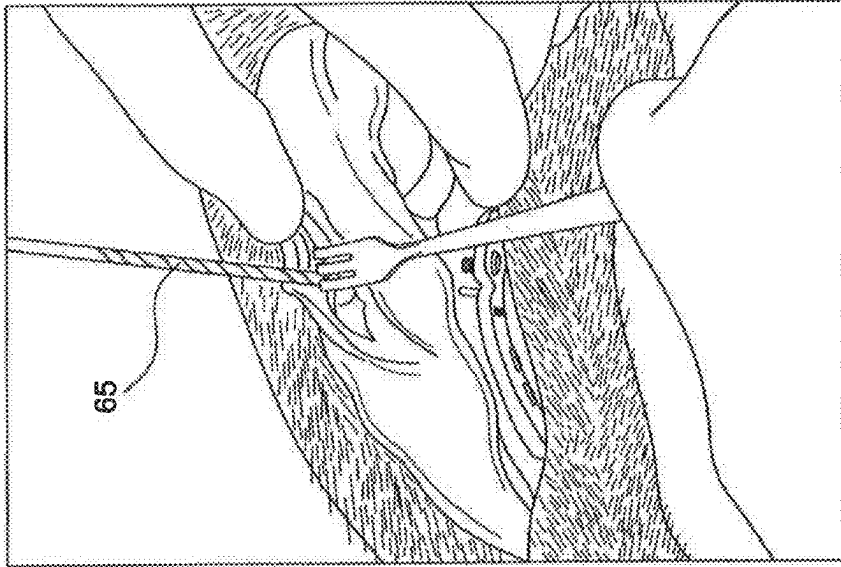


FIG. 23(a)

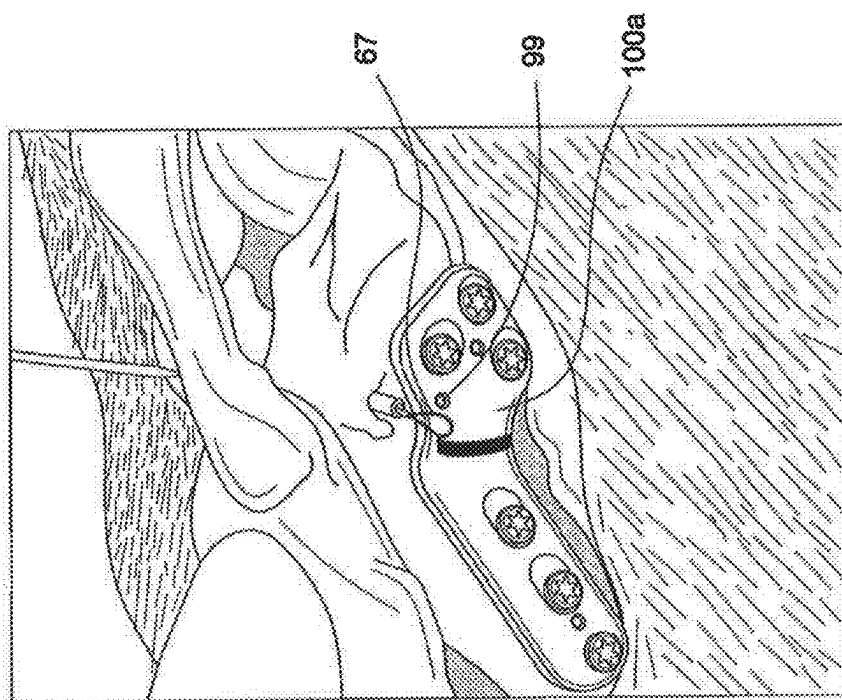


FIG. 24

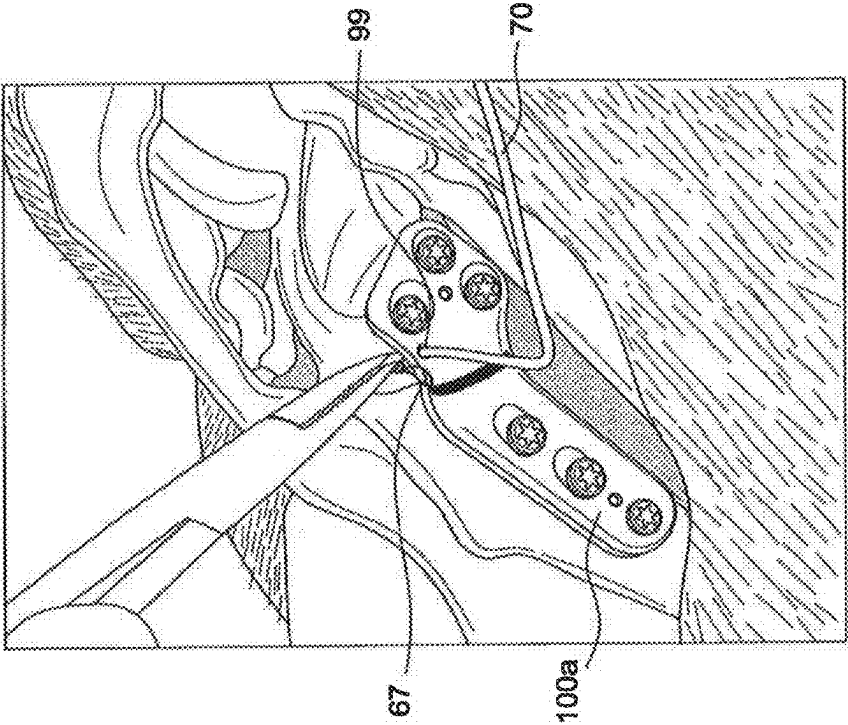


FIG. 25

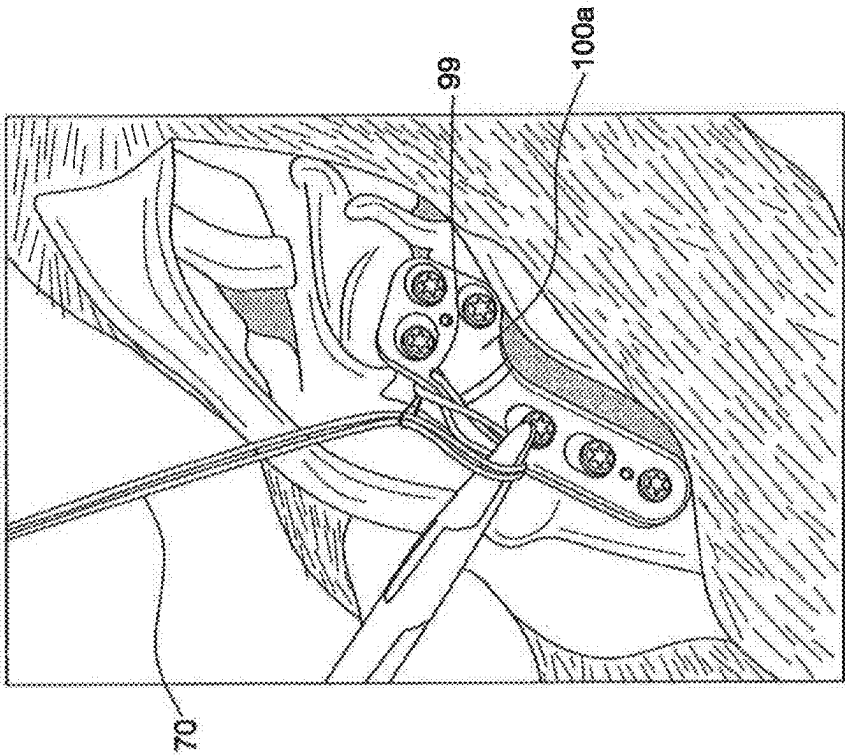


FIG. 26

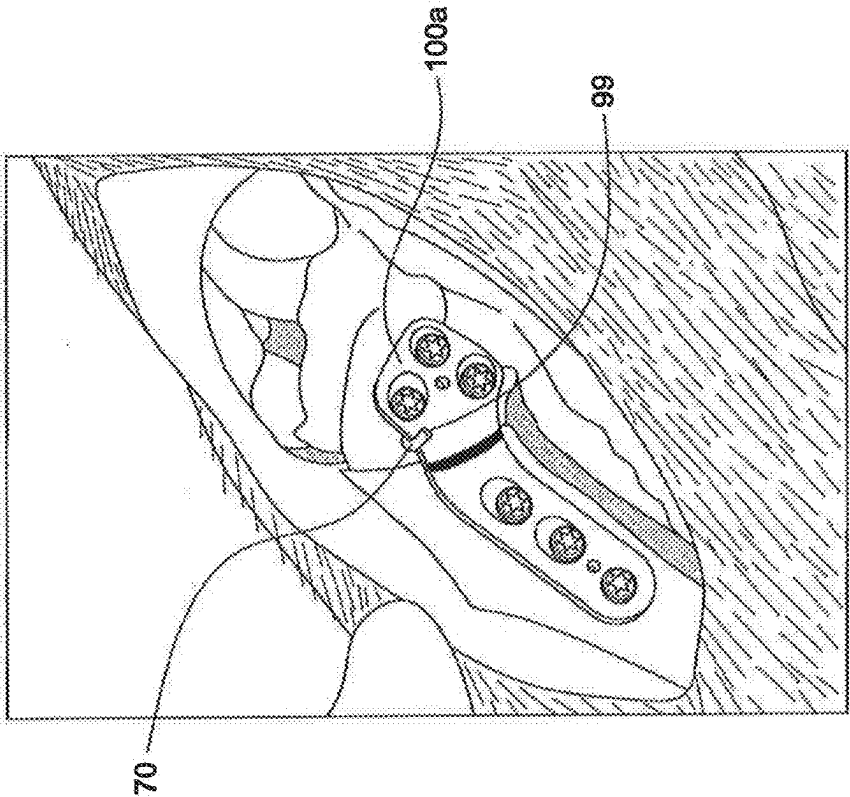
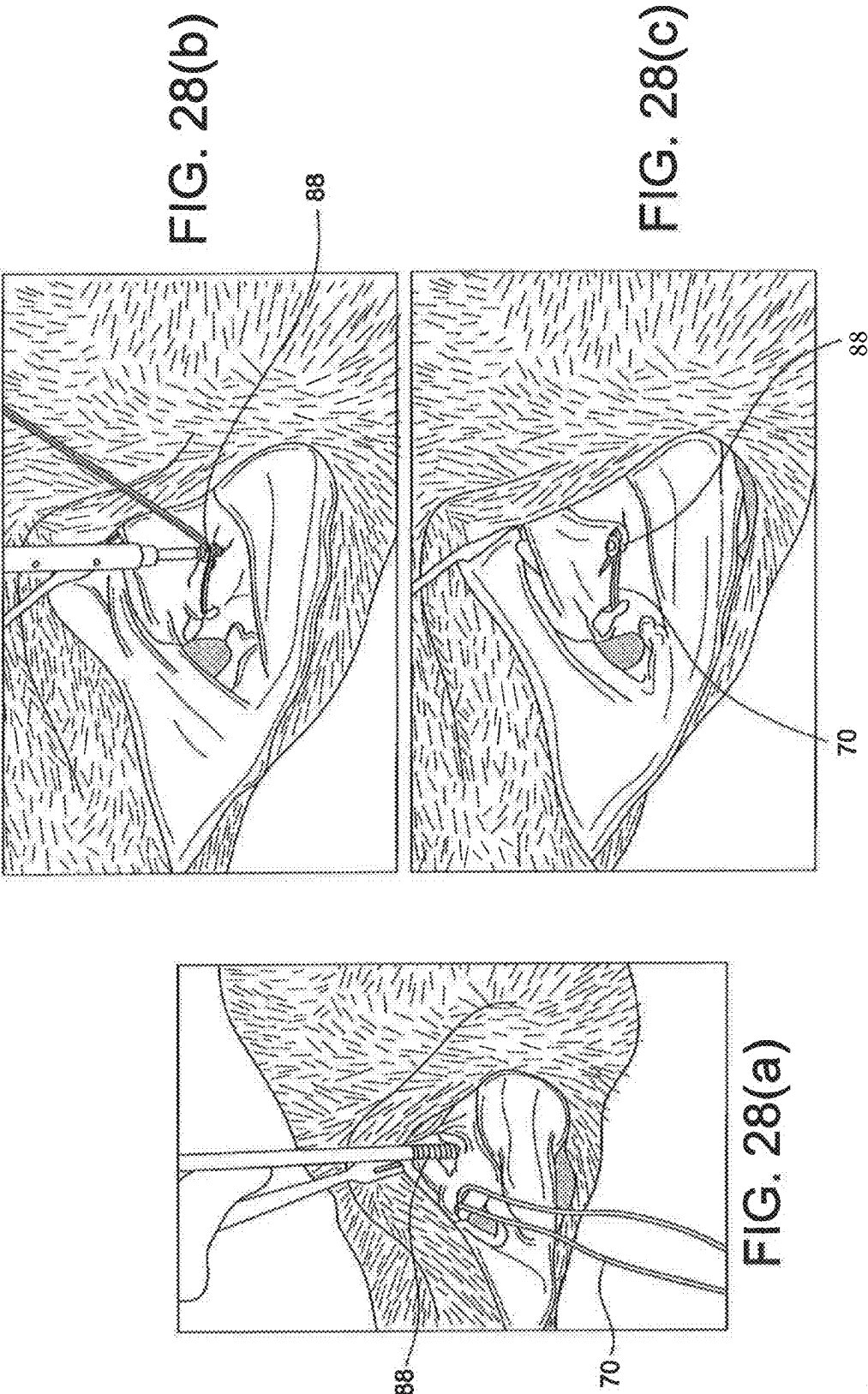


FIG. 27



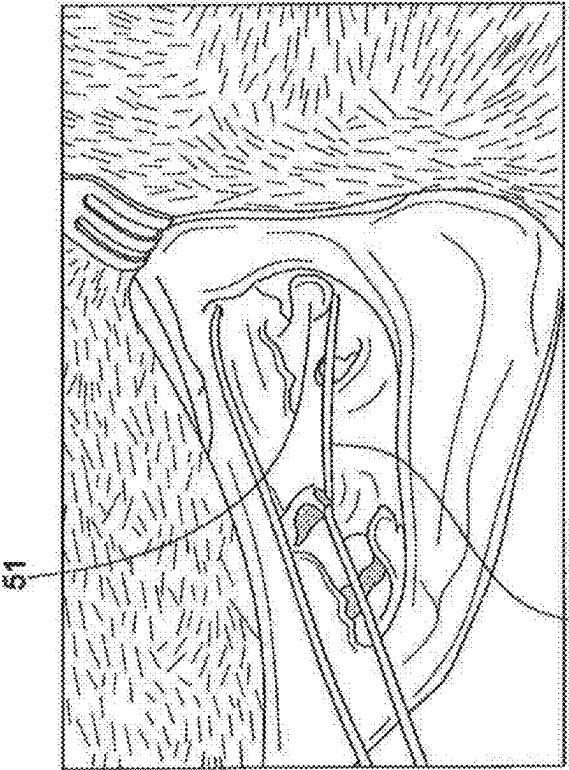


FIG. 29(a)

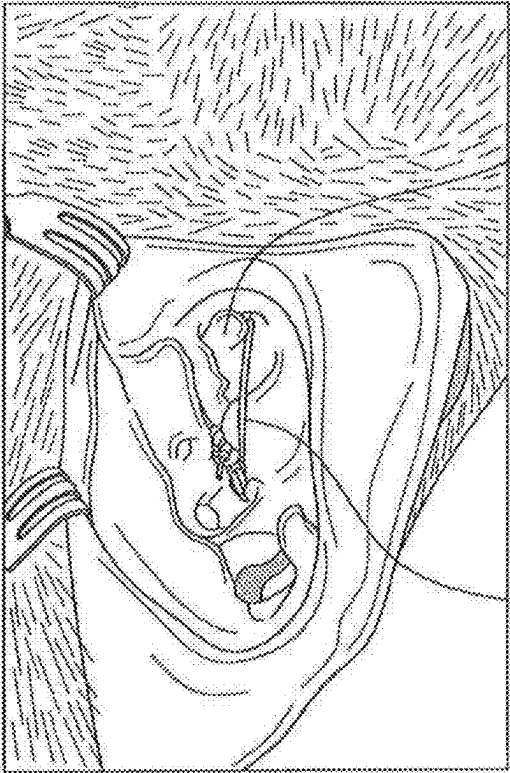
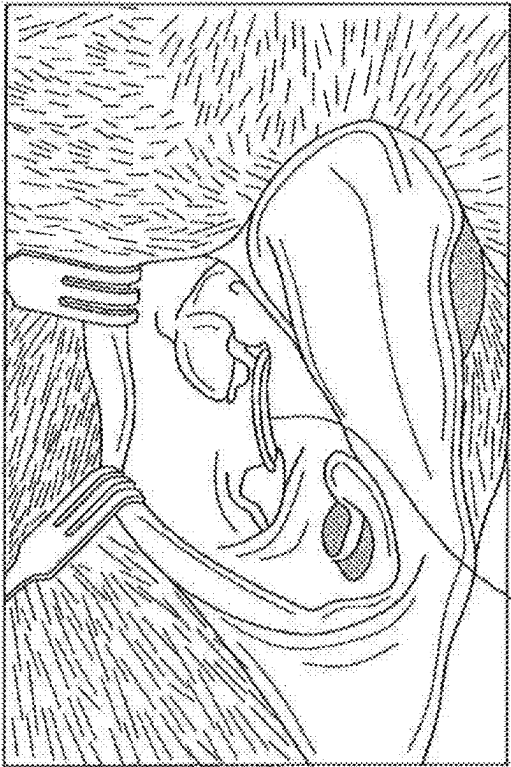
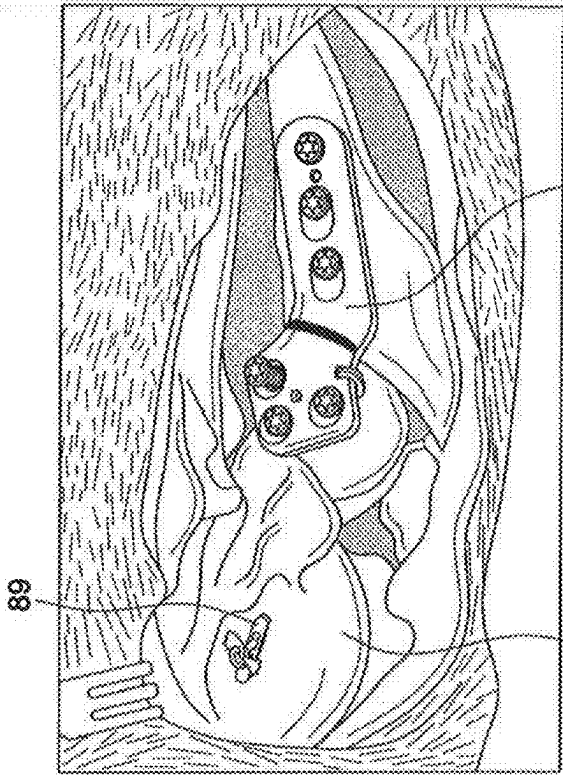


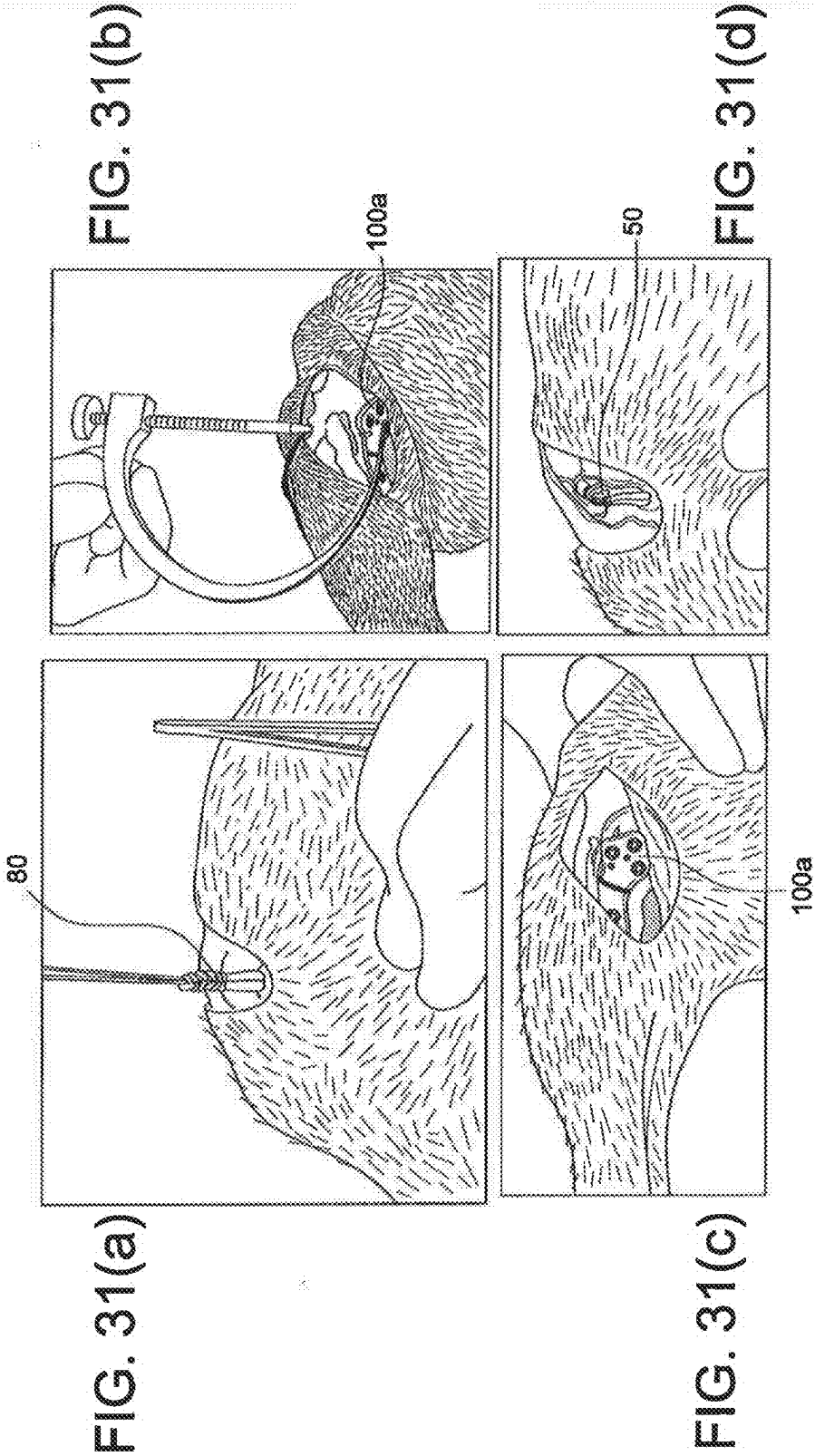
FIG. 29(b)



70 FIG. 30(a)



89 53 100a FIG. 30(b)



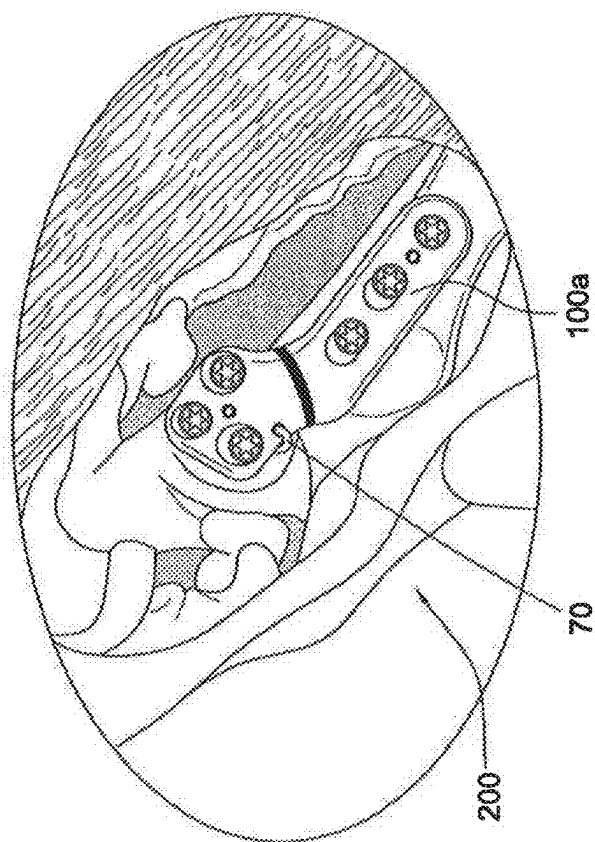


FIG. 32

TIBIAL PLATEAU LEVELING OSTEOTOMY PLATE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation of U.S. patent application Ser. No. 17/858,123, filed on Jul. 6, 2022, which is a continuation of U.S. patent application Ser. No. 17/672,873, filed on Feb. 16, 2022 (now U.S. Pat. No. 11,419,648), which is a continuation of U.S. patent application Ser. No. 17/348,888, filed on Jun. 16, 2021 (now U.S. Pat. No. 11,298,167), which is a continuation of U.S. patent application Ser. No. 16/420,602, filed on May 23, 2019 (now U.S. Pat. No. 11,096,729), which is a continuation of U.S. patent application Ser. No. 14/533,768, filed on Nov. 5, 2014 (now U.S. Pat. No. 10,299,841), which claims the benefit of U.S. Provisional Application No. 61/900, 123, filed on Nov. 5, 2013, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to methods and apparatus for surgical procedures.

BACKGROUND OF THE INVENTION

[0003] Tibial plateau leveling osteotomy (TPLO) procedures are well known in the veterinary art. Tibial plateau leveling osteotomy procedures are used to correct ruptured cranial cruciate ligaments for various animals, particularly canines. These procedures provide an alternative therapy to ligament repair procedures. Tibial plateau leveling osteotomy procedures have become the standard of care for medium and large canines.

[0004] When the canine ligament partially or fully ruptures, the tibial plateau leveling osteotomy procedure provides a way to correct this problem. A curvilinear cut is made to the upper portion of the tibia. This cut portion of the tibia is then rotated on the order of about 20-30 degrees to create a more level plane or surface on the top of the tibia upon which the femur can rest. The cut and repositioned portion of the tibia is then secured to the lower portion of the tibia.

[0005] There is a need for a TPLO bone plate provided with a new design that allows surgeons to secure to one or more tibial segments of an animal, as part of a tibial leveling osteotomy procedure for an animal. Also needed is a TPLO bone plate that offers a means for providing rotational stability, particularly means for providing additional rotational stability to the TPLO procedure. Also needed is a method of fracture or osteotomy repair in animals that allows securing of dissected soft tissue to the anatomic position where it was dissected, during the fracture or osteotomy repair.

BRIEF SUMMARY OF THE INVENTION

[0006] The present invention provides apparatus and methods for bone-tissue fixation (for example, bone-soft tissue fixation) in animals, particularly canines, using a TPLO plate having one or more suture eyelets/holes/slots provided within the body of the plate to allow the user (surgeon) to confer means for providing rotational stability to the joint or the bone, and to the final TPLO repair. The TPLO plate may be also employed, if necessary, to reattach tissue (bone or soft tissue, for example) to the plate at the

anatomical location where the tissue was dissected. The suture eyelets/holes/slots may be provided in any number and at any location on the plate, for example, on the proximal region of the plate.

[0007] The TPLO plate of the present invention includes a body preferably formed of a metal and one or more suture eyelets (suture holes or slots) or other similar features which are incorporated into the shaft (for example, the proximal region) of the plate. The suture eyelets (holes) may have various shapes, forms and configurations and may be provided on or within a surface of the bone plate in any number, depending on the characteristics of the osteotomy or of the plate design. The suture eyelets preferably receive a flexible strand for fixation of tissue (bone or soft tissue) to the bone plate to improve the rotational stability of the bone or joint.

[0008] The present invention also provides a method of improving the rotational stability of a joint or bone, for example, of the knee or stifle during a TPLO procedure. The method comprises the steps of: (i) providing a TPLO plate that includes one or more suture eyelets/holes/openings/slots on or within the bone plate; (ii) fixing the TPLO plate to bone with fasteners such as screws; (iii) passing suture through one or more suture eyelets of the TPLO plate; and (iv) securing the suture to bone to improve the rotational stability of the bone/joint and the overall TPLO repair.

[0009] These and other features and advantages of the invention will be more apparent from the following detailed description that is provided in connection with the accompanying drawings and illustrated exemplary embodiments of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1-3 illustrate schematic subsequent method steps of a TPLO procedure with an exemplary TPLO bone plate of the present invention.

[0011] FIGS. 4-6 illustrate various X-ray views of a TPLO plate of the prior art positioned on canine tibia.

[0012] FIGS. 7-9 illustrate various X-ray views of a TPLO plate of the present invention (superimposed on the TPLO plate of FIGS. 4-6) to illustrate the differences between the TPLO plate of the present invention and the prior art TPLO plate.

[0013] FIGS. 10(a)-(d) illustrate various views of an exemplary 2.7 mm standard TPLO plate of the present invention.

[0014] FIGS. 11(a)-(d) illustrate various views of an exemplary 3.5 mm small TPLO plate of the present invention.

[0015] FIGS. 12(a)-(d) illustrate various views of an exemplary 3.5 mm standard TPLO plate of the present invention.

[0016] FIGS. 13(a)-(d) illustrate various views of an exemplary 3.5 mm broad TPLO plate of the present invention.

[0017] FIGS. 14(a)-(d) illustrate various views of an exemplary 4.5 mm standard TPLO plate of the present invention.

[0018] FIGS. 15-32 illustrate subsequent steps of an exemplary method of TPLO bone plate fixation according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0019] The present invention provides TPLO bone plates having one or more suture eyelets on or within the plate, to allow the user (surgeon) to increase the rotational stability of the bone or joint and, optionally, to reattach the soft tissue to the plate at the anatomical location where the tissue was dissected.

[0020] The TPLO bone plates of the present invention are provided in various sizes, for example, 2.7 mm standard, 3.5 mm small, 3.5 mm standard, 3.5 mm broad, 4.5 mm standard, etc., accommodating correspondingly-sized screws. The TPLO bone plates are designed to accept various locking and non-locking low profile cortical screws. In an exemplary embodiment, each plate is provided with a suture hole located cranially on the proximal portion of the plate that accepts one flexible strand, for example a strand of #5 FiberWire® or FiberTape®. Each plate is provided with a surface cut demonstrating where a laser line for the osteotomy will be. All angles and radii for the proximal contour may be equal for each plate design. Some geometries between the 2.7 mm, 3.5 mm, 4.5 mm plates are slightly different.

[0021] The TPLO plates of the present invention may be provided as single devices and/or as part of kits incorporating additional instruments and devices. For example, the TPLO plates may be provided with additional components known for TPLO procedures, such as locking and non-locking screws, threaded drill guide, drill guide, graduated drill bit, screwdriver such as hexalobe screwdriver, Quick-Connect driver such as hexalobe QuickConnect driver, all sized according to the particular dimensions of the TPLO plate to be employed with.

[0022] The present invention also provides a TPLO procedure that improves the rotational stability of the joint or bone. The method comprises the steps of: (i) providing a TPLO plate that includes one or more suture eyelets/holes/openings/slots/apertures on or within the bone plate; (ii) fixating the TPLO plate to bone with fasteners such as screws; (iii) passing suture through one or more suture eyelets of the TPLO plate; and (iv) tying the suture to bone to improve the rotational stability of the joint or bone.

[0023] An exemplary method of fixation of anatomical tissue during surgical applications by employing a TPLO plate that improves the rotational stability of the bone or joint or repair site (surgical site), and also allows reattachment of soft tissue to bone and to the plate, comprises the steps of: (i) providing a TPLO plate that includes one or more suture eyelets/holes/openings/slots/apertures on or within the bone plate; (ii) placing the TPLO plate on the bone and dissecting the adjacent soft tissue to allow the plate to fit; (iii) fixating the TPLO plate to bone with fasteners such as screws; (iv) passing suture around the bone and through the suture eyelets of the TPLO plate; (v) passing suture through the dissected soft tissue and through the suture eyelets of the TPLO plate; and (vi) tying the suture to improve the rotational stability of the bone/joint/repair site, and to secure the dissected soft tissue to the anatomical position/location where it was dissected.

[0024] Referring now to the drawings, where like elements are designated by like reference numerals, FIGS. 1-3 illustrate a TPLO plate 100 of the present invention provided with a plurality of suture eyelets 99 (openings/slots/holes/apertures 99) incorporated into the plate to allow soft tissue

attachment to the plate. For exemplary purposes only, the TPLO plate 100 is illustrated and described below as a bone plate for providing improved rotational stability in osteotomies of an animal such as a canine, and attachment of dissected biceps and/or adjacent muscles. However, the invention has applicability to the fixation of other bones or bone segments of canines or of other animals, including the fixation of associated soft tissue to bones or bone segments.

[0025] As shown in FIG. 2, the TPLO plate 100 includes a rigid body 10 with a longitudinal axis 11, a transversal axis 13, a first surface 17 and a second bone-contacting surface 19. Body 10 is preferably formed of a metal such as titanium, titanium alloy, stainless steel, or other materials that possess mechanical and physical properties suitable for coupling bones together. Body 10 is also defined by a distal region 12 (shaft 12) surrounded by a distal edge 12a and a proximal region 14 surrounded by a proximal edge 14a.

[0026] The body 10 of the TPLO plate 100 is provided with a first plurality of through holes or openings 55a, 55b that receive corresponding screws or similar fixation devices 25 to secure the bone plate to bone. The fixation devices 25 may be screws, anchors, washers or any combination thereof for securing the bone plate to bone. The first plurality of openings 55a, 55b extend from the first surface 17 through the body 10 and to the bone-contacting surface 19, for accommodating at least one fixation device. Openings 55a, 55b may be provided in any number and may have similar or different perimeters. Openings 55a, 55b may be also optimally placed in the body 10 of the plate and at various angles with respect to transversal axis 13 of the plate 100. Additional small holes may be added within the body of the plate for the purpose of inserting k-wires to help keep the plate stable during the process of fixating the plate to the bone. Exemplary plate designs may incorporate a single suture hole with a diameter approximately double than that of the diameter of k-wire holes.

[0027] The first plurality of openings 55a, 55b includes a first set of holes 55a located within the proximal region 14 and the distal region 12 and extending asymmetrically along the longitudinal axis 11 of the body 10, and a second set of holes 55b located only within the distal region 12 and in asymmetry relative to the longitudinal axis 11 of the body 10.

[0028] At least one of the openings 55a may include spherical bushings to allow variable angle-locking to be achieved by threading at least one tapered locking screw into the spherical bushings, as detailed below. The tapered head will expand the bushing, locking the screw to the plate. Non-locking screws may be also employed and can be placed in any opening 55a. If polyaxial bushings is employed, the polyaxial bushings offer the surgeon the freedom to direct the fixation devices (for example, screws) within the anatomical template, based on the fracture pattern and bone quality for better fixation and to achieve a low profile polyaxial suture plate with any angulation, for example, a 60 degree angulation. The polyaxial bushings in each hole provide multi-directional locking or non-locking capability of the plate/screw construct to the bone. Drill guides can angulate the bushing to the desired screw trajectory.

[0029] Openings 55a may be provided in any number and may have similar or different perimeters. In the exemplary embodiment shown in FIG. 2, openings 55a are circular and have the same diameter, while openings 55b are oval and

have same shape and dimensions. Openings **55a**, **55b** may be also optimally placed in the body **10** of the bone plate and at various angles with respect to a transversal axis of the TPLO plate **100**. Openings **55a**, **55b** are preferably arranged non-linearly with respect to the longitudinal axis **11**, for the 3.5 mm broad plates and 4.5 standard plates. In addition, as and shown in FIG. 2, axis **11a** passing through both centers of openings **55a** located on proximal region **14** and disposed along edge **14a** is about parallel to the longitudinal axis **11** of the body of the plate **100**.

[0030] As also shown in FIGS. 1-3, the TPLO plate **100** includes a second plurality of through holes or apertures including one or more suture eyelets **99** (suture holes or slots) which are disposed within the body **10** of the bone plate **100**. Exemplary TPLO plate **100** includes one, two or three suture eyelets or holes **99**.

[0031] The suture eyelets **99** preferably receive a flexible strand **70** (FIG. 3) to increase the rotational stability of the bone or joint. The suture eyelets may also help in the fixation of soft tissue **80** to the bone plate **100**. For example, and if desired, suture eyelets **99** allow the user (surgeon) to reattach soft tissue **80** to the plate at the anatomical location where the tissue was dissected. The suture eyelets **99** may have various shapes, forms and configurations and may be provided on or within a surface of the bone plate **100** in any number, depending on the characteristics of the osteotomy or of the plate design.

[0032] Suture eyelets **99** may be recessed and they extend from the first surface **17** through the body **10** and to the bone-contacting surface **19**, to accommodate at least one flexible strand attached to soft tissue to be reattached to the bone and to increase the rotational stability of the bone or joint. Although reference to the eyelets **99** will be made below as to the suture eyelets **99**, the invention is not limited to this exemplary-only embodiment and contemplates eyelets for passing of any flexible strand, for example, suture, suture tape such as FiberTape®, suture chain such as FiberChain®, or a flexible material forming (or part of) a continuous loop/button construct provided with a button and a continuous loop attached to the button, among many others.

[0033] Suture eyelets **99** may be chamfered suture eyelets or provided with fillets. Suture eyelets **99** are provided within the body of the plate **100** to allow the user (surgeon) to pass one or more flexible strands (for example, one or more suture strands such as FiberWire® suture) after or before the plate is fixed to bone. In this manner, suture eyelets **99** allow the user (surgeon) to reattach soft tissue **80** to the plate at the anatomical location where the tissue was dissected and to also increase the rotational stability of the bone or joint. The suture eyelets **99** (holes/apertures/openings or chamfered suture eyelets **99**) may have various shapes, forms and configurations and may be provided on or within a surface of the TPLO plate **100** in any number. The suture eyelets **99** preferably receive at least one flexible strand **70** for fixation of soft tissue **80** to the TPLO plate **100**.

[0034] One or more flexible strands may be passed through one suture eyelet. One flexible strand may be passed through one or more suture eyelets.

[0035] In an exemplary embodiment, the first plurality of openings **55a** have a first uniform diameter “D” (FIG. 2) and the second plurality of suture eyelets **99** have a second uniform diameter “d” (FIG. 2) which is smaller than the first diameter “D.” The first diameter “D” is preferably at least four times larger than the second diameter “d.”

[0036] According to another exemplary embodiment, the bone plate **100** of the present invention has one suture eyelet and two other small holes that are intended for passing k-wires or similar instruments to facilitate stability of the plate prior to fixation with screws. The holes of the plate for accommodating fixation devices (screws) preferably are provided asymmetrically relative to the longitudinal axis of the body of the plate. Other embodiments may also have only one oval hole or two oval holes that are centered on the longitudinal axis of the body of the plate.

[0037] The present invention also provides a suture plate kit (assembly) with a bone plate **100** (suture plate **100**) of the present invention and a suturing kit including at least one flexible strand attached to at least one needle. The at least one flexible strand may be FiberWire® suture containing color coded suture, for example, a #2 and/or #5 FiberWire® sutures, with needles on each end, to facilitate suturing the soft tissue to the bone plate after the plate is in place. The needles may be attached to the suture by any method known in the art, for example, by being swedged onto the suture. FIG. 3 illustrates exemplary TPLO bone plate **100** of the present invention with tissue **80** to be attached to its proximal region **14** and with attached suture **70** to be passed through one of the suture eyelets **99** (the cranial proximal eyelet **99**) for further attachment of soft tissue to TPLO plate **100**.

[0038] The present invention also provides a method of forming a TPLO plate having suture holes for soft tissue attachment (or reattachment) for surgical application and for increasing the rotational stability of the joint or the bone. One to three suture eyelets **99** are formed within the body of a TPLO plate to enable a surgeon to easily place the tissue **80** to be attached/reattached back to the bone (for example, ligament or other soft tissue) during a TPLO procedure.

[0039] The present invention also provides a method of stabilizing a canine proximal tibia, comprising inter alia the steps of: (i) providing a TPLO plate having a body with a first surface, a second surface opposed to the first surface, the body having a proximal region and a distal region, a first plurality of apertures passing through the plate, the first plurality of apertures accommodating a plurality of fixation devices, a second plurality of apertures passing through the plate, the second plurality of apertures accommodating a plurality of flexible strands attached to soft tissue to be reattached to the bone; (ii) securing the TPLO plate to tibia by inserting the plurality of fixation devices at various angles relative to a transversal axis of the body; (iii) passing at least one flexible strand (for example, monofilament suture) through at least one of the second plurality of apertures; and (iv) securing the at least one flexible strand to femur or to soft tissue with a fixation point on the femur, to improve the rotation stability of canine bone or joint.

[0040] FIGS. 1-3 illustrate exemplary steps of a TPLO procedure with TPLO plate **100** of the present invention for repair of bone segments of tibia **50** (at a tibia-femur joint region in the leg of an exemplary canine).

[0041] FIG. 1: Use a current C-guide **60** or similar instrument to allow driving suture passing pin **66** (FIG. 3) through plate holes to desired tibial tunnel entrance.

[0042] FIG. 2: The TPLO plate **100** has two or three exemplary suture holes/slots **99** at a location where they would work. Could also have just one or button recess, depending on the characteristics of the particular repair.

[0043] FIG. 3: Prepare biceps tendon **80** for transposition for bio-extracorporeal; pull end of suture **70** (one at a time through, to tie over the TPLO plate **100**). Could also repeat procedure with fabellar suture, anchor, or suture loop/fixation device (for example, a TightRope® device), as detailed below. This procedure would also eliminate the need to extend the incision or struggle to reach to the lateral femoral area.

[0044] FIGS. 4-9 illustrate various views of the shape and position holes of a prior art plate **1** and of an exemplary TPLO plate **100'** of the present invention.

[0045] FIG. 5: plate **1** is tilted to position; proximal screws are located in the middle of the fragment.

[0046] FIG. 6: plate **1** is straight but the screws are too close with the osteotomy cranially.

[0047] FIG. 7: illustrates exemplary TPLO plate **100'** of the present invention superimposed with plate **1** of the prior art.

[0048] FIG. 8: illustrates TPLO plate **100'** of FIG. 7 with an exemplary eyelet **99a** located at a first position on the proximal region of the body of the plate (Option 1).

[0049] FIG. 9: illustrates TPLO plate **100'** of FIG. 7 with another exemplary eyelet **99b** located at a second position on the proximal region of the body of the plate (Option 2).

[0050] FIGS. 10(a)-(d) illustrate various views of an exemplary 2.7 mm standard TPLO plate **100a** of the present invention. FIGS. 11(a)-(d) illustrate various views of an exemplary 3.5 mm small TPLO plate **100b** of the present invention. FIGS. 12(a)-(d) illustrate various views of an exemplary 3.5 mm standard TPLO plate **100c** of the present invention. FIGS. 13(a)-(d) illustrate various views of an exemplary 3.5 mm broad TPLO plate **100d** of the present invention. FIGS. 14(a)-(d) illustrate various views of an exemplary 4.5 mm standard TPLO plate **100e** of the present invention.

[0051] Reference is now made to FIGS. 15-32 which illustrate subsequent steps of a method of TPLO bone plate fixation according to an embodiment of the present invention and employing an exemplary TPLO plate **100**, **100a**, **100b**, **100c**, **100d**, **100e**, **100'** of the present invention.

[0052] The method begins by fixating exemplary TPLO plate **100a** by employing known fixation steps in the art and as listed below:

[0053] Create and pin the osteotomy **55** at the level of tibia **50** (FIG. 15);

[0054] Check the TPLO plate **100a** first;

[0055] Position TPLO plate **100a** using K-wires **56** (for example, 0.062" K-wires **56**) (FIG. 16);

[0056] Fixate TPLO plate using a fixation instrument (for example, a push-pull device, or a clamp/plier type device to fixate the plate);

[0057] Drill in loaded position of compression slot (for example, proximal compression slot);

[0058] Check screw depth;

[0059] Insert non-locking screw partially;

[0060] Drill proximal holes **15** and insert screws **25** (FIG. 17);

[0061] Remove proximal positioning K-wire **56** (FIG. 18);

[0062] Remove osteotomy K-wire;

[0063] Drill and insert final proximal screw **25** (FIG. 19);

[0064] Remove distal K-wire;

[0065] Fully insert compression screw;

[0066] Drill and insert distal locking screw **25** (FIG. 20);

[0067] Remove fixation instrument (for example, push-pull device, or clamp/plier type device);

[0068] Drill and insert screw in compression slot (for example, distal compression slot) (FIG. 21).

[0069] FIG. 22: align C-Ring Guide **60** with suture hole **99** and drill K-wire;

[0070] FIGS. 23(a) and 23(b): use 2.7 mm drill **65** over the K-wire;

[0071] FIG. 24: feed suture passer **67** through the tunnel;

[0072] FIG. 25: pass a flexible strand **70** for example, #5 FiberWire® suture **70**, through the suture hole **99**;

[0073] FIG. 26: pass tails of the #5 FiberWire® **70** through the nitinol loop **67**;

[0074] FIG. 27: pass suture **70** through the tunnel;

[0075] FIGS. 28(a)-(c): fixate the suture **70**—first method: use a knotless fixation device **88** such as SwiveLock® **88**;

[0076] FIGS. 29(a) and 29(b): fixate the suture **70**—second method: tie suture **70** around fabella **51**;

[0077] FIGS. 30(a) and 30(b): fixate the suture **70**—third method: use a button **89** on the medial side of femur **53** and attach suture **70** to the button **89**;

[0078] FIGS. 31(a)-(d): fixate the suture **70**—fourth method: use suture holes **99** for fixation of tendons, for example, biceps **80**.

[0079] FIG. 32 illustrates a final exemplary TPLO repair **200** of the present invention with exemplary TPLO plate **100a**.

[0080] The at least one flexible strand **70** may be suture or a high-strength suture, such as FiberWire® suture, which is disclosed and claimed in U.S. Pat. No. 6,716,234, the entire disclosure of which is incorporated by reference in its entirety herein. FiberWire® suture is formed of an advanced, high-strength fiber material, namely ultrahigh molecular weight polyethylene (UHMWPE), sold under the tradenames Spectra (Honeywell) and Dyneema (DSM), braided with at least one other fiber, natural or synthetic, to form lengths of suture material. The at least one flexible strand **70** may be also suture tape such as FiberTape®, suture chain such as FiberChain®, or a flexible material that is part of a continuous loop/button construct provided with a button and a continuous loop attached to the button. The flexible strand **70** may be also a monofilament, as this works well in veterinarian applications.

[0081] In another exemplary embodiment, the flexible strand **70** may be a continuous loop formed of a plurality of suture strands configured to separate from a single strand to a plurality of strands in a continuous loop. In yet another embodiment, the flexible strand is an adjustable loop (forming a TightRope® ACL construct) which consists of two interconnected, adjustable flexible loops formed by splicing a suture strand in a manner disclosed in U.S. Pat. No. 8,460,379 issued on Jun. 11, 2013 and U.S. Pat. No. 8,439,976 issued on May 14, 2013, the disclosures of both of which are incorporated by reference herein in their entireties.

[0082] The flexible strand **70** may be also part of a suture loop/needle construct similar to the FiberLoop® construct detailed and disclosed in U.S. Pat. No. 8,298,284 issued on Oct. 30, 2012, the disclosure of which is incorporated by reference herein in its entirety. The flexible strand may be suture tape such as FiberTape® (as disclosed in U.S. Pat. No. 7,892,256) or collagen tape, or combinations thereof.

[0083] The fixation devices/implants **25, 88** may have various sizes (various diameters and/or lengths) and may be formed of biocompatible materials such as PEEK, biocomposite materials, metals and/or metal alloys, or combination of such materials, among others. The fixation devices/implants **25, 88** may be also formed of any rigid medically approved materials, for example, plastic or carbon fiber, or combination of different materials.

[0084] While the present invention is described herein with reference to illustrative embodiments for particular applications, it should be understood that the invention is not limited thereto. Those having ordinary skill in the art and access to the teachings provided herein will recognize additional modifications, applications, embodiments and substitution of equivalents all fall within the scope of the invention.

What is claimed is:

1. A surgical apparatus comprising:
 - a tibial plateau leveling osteotomy (TPLO) plate including a body that includes a proximal region, a distal region, a first surface, and a bone contacting surface opposite the first surface,
 - wherein the proximal region is angled relative to the distal region such that the proximal region is offset from either a left-hand side edge or a right-hand side edge of the distal region,
 - wherein, for a right-hand side offset, both a left-hand side edge and a right-hand side edge of the proximal region are shifted to the right of the left-hand side edge of the distal region, or for a left-hand side offset, both the left-hand side edge and the right-hand side edge of the proximal region are shifted to the left of the right-hand side edge of the distal region; and
 - a first plurality of apertures passing through the body and each being configured to accommodate a fixation device;
 - wherein a longitudinal axis of the body bisects a majority of a length of the distal region and intersects through a first portion of the proximal region,
 - wherein the longitudinal axis intersects a first aperture of a first portion of the first plurality of apertures that are disposed in the proximal region,
 - wherein a second axis intersects each of a second aperture and a third aperture of the first portion of the first plurality of apertures,
 - wherein the second axis extends at an acute angle relative to the longitudinal axis,
 - wherein the proximal region extends in a distal-to-proximal direction away from the distal region along a centerline axis that bisects the proximal region into two equal parts, and further wherein the centerline axis intersects a first curved peak of the proximal region,
 - wherein the proximal region is a proximal end region establishing a tibial plateau leveling head portion of the TPLO plate, and the distal region is a distal shaft region establishing a shaft portion of the TPLO plate.
2. The surgical apparatus as recited in claim 1, wherein the first plurality of apertures includes a combination of circular holes and oval holes.
3. The surgical apparatus as recited in claim 1, wherein the body of the TPLO plate is a rigid body comprised of a metallic material.
4. The surgical apparatus as recited in claim 3, wherein the metallic material is stainless steel.

5. The surgical apparatus as recited in claim 1, comprising a second plurality of apertures passing through the body and each configured to accommodate a flexible strand or a k-wire, wherein at least one of the second plurality of apertures is disposed in each of the proximal region and the distal region of the body.

6. The surgical apparatus as recited in claim 1, wherein the second axis intersects through a second portion of the proximal region that is near a second curved peak and through a third portion of the proximal region that is near a third curved peak of the proximal region.

7. The surgical apparatus as recited in claim 1, wherein the distal region protrudes outwardly to a curved distal peak at a single location of the distal region.

8. The surgical apparatus as recited in claim 1, wherein the body of the TPLO plate includes a neck region disposed between the proximal region and the distal region, and further comprising a depression formed in the neck region.

9. The surgical apparatus as recited in claim 1, wherein the first aperture is formed adjacent to a second curved peak of the proximal region, and further wherein the first curved peak is disposed nearer to the centerline axis of the proximal region than either the second curved peak or a third curved peak of the proximal region.

10. The surgical apparatus as recited in claim 9, wherein the first curved peak is located between the second curved peak and the third curved peak.

11. The surgical apparatus as recited in claim 9, wherein the first curved peak establishes a proximal-most tip of the proximal region.

12. The surgical apparatus as recited in claim 9, wherein the first curved peak is disposed further away from the distal region than the second curved peak of the proximal region.

13. The surgical apparatus as recited in claim 9, wherein the second aperture is formed adjacent to the first curved peak of the proximal region and the third aperture is formed adjacent to the third curved peak of the proximal region.

14. The surgical apparatus as recited in claim 1, wherein the first aperture, the second aperture, and the third aperture are substantially equally sized apertures.

15. The surgical apparatus as recited in claim 1, wherein the proximal region is acutely angled relative to the distal region, and further wherein the first portion of the proximal region through which the longitudinal axis intersects is located at the left-hand side edge or the right-hand side edge of the proximal region.

16. The surgical apparatus as recited in claim 1, wherein the second axis is closer to parallel than perpendicular relative to the longitudinal axis.

17. The surgical apparatus as recited in claim 1, wherein a first portion of a length of the body is established by the distal region and a second portion of the length of the body is established by the proximal region, wherein the first portion of the length is larger than the second portion of the length.

18. The surgical apparatus as recited in claim 1, wherein the first aperture is formed adjacent to a second curved peak of the proximal region and the second aperture is formed adjacent to the first curved peak of the proximal region.

19. The surgical apparatus as recited in claim 18, wherein the centerline axis intersects through the second aperture.

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